

02 — Dividers, source transformation, ladders

Class notes p.1, p.4–5 · 4 board questions · needs 01

Map

1 Voltage division

→

2 Current division

→

3 Division with AC

→

4 Source transformation

→

5 Ladder reduction

The questions this file answers

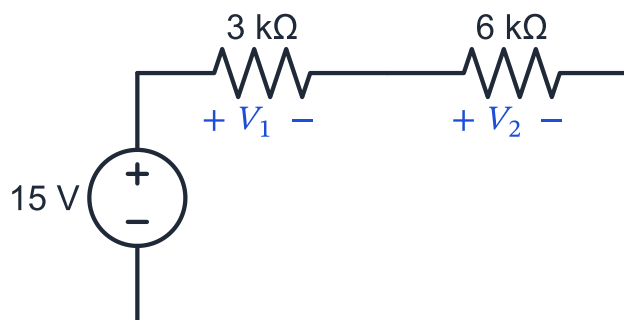
- **Class p.4** 15 V across 3 kΩ and 6 kΩ in series. Find V_1 and V_2 .
- **Class p.5** A 12 A source feeds 3 kΩ || 6 kΩ || 4 kΩ. Find the current in each.
- **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 kΩ and 7 kΩ in series. Find v_1 and v_2 .
- **Class p.1** An infinite ladder: 3 kΩ shunt, 5 kΩ series, 7 kΩ shunt, 11 kΩ series, 13 kΩ shunt, ... Find R_{eq} .

Build

1 · Voltage division

CLASS P.4

15 V is applied across 3 kΩ and 6 kΩ in series. Find the voltage across each.



Series resistors carry the same current, so each one's voltage is proportional to its resistance.

GUESS FIRST

Which resistor gets more voltage, and roughly how much?

Series means **one current**: $I = \frac{V}{R_1 + R_2}$. Each voltage is that current times its own resistance:

$$V_1 = V \cdot \frac{R_1}{R_1 + R_2} \quad V_2 = V \cdot \frac{R_2}{R_1 + R_2}$$

$V_1 = 15 \times \frac{3}{9} = 5 \text{ V}$, $V_2 = 15 \times \frac{6}{9} = 10 \text{ V}$. They add back to 15 V (KVL).

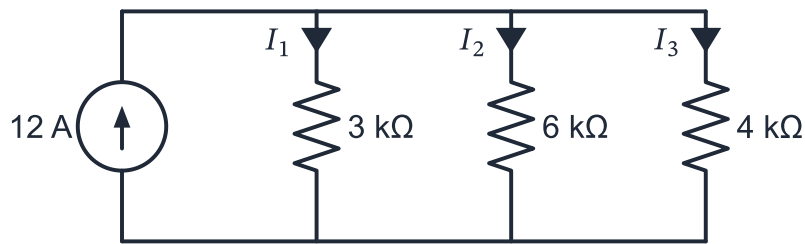
CHECK 1

12 V across 1 k Ω , 2 k Ω , 3 k Ω in series. Voltage across the 2 k Ω ?

2 · Current division

CLASS P.5

A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .



Parallel branches share one voltage, so the current splits in inverse proportion to resistance.

GUESS FIRST

Which branch carries the most current? Rank the three before calculating.

Parallel means **one voltage**. For two branches that gives the class rule, where **each branch gets the other resistor on top**:

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2} \quad I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

For three branches, the professor's method is to merge the other two branches into one, then apply the two-branch rule:

Branch	The other two, merged	Current
3 k Ω	$6 \parallel 4 = 2.4 \text{ k}\Omega$	$12 \times \frac{2.4}{2.4 + 3} = 5.33 \text{ A}$
6 k Ω	$3 \parallel 4 = \frac{12}{7} \text{ k}\Omega$	$12 \times \frac{12/7}{12/7 + 6} = 2.67 \text{ A}$
4 k Ω	$3 \parallel 6 = 2 \text{ k}\Omega$	$12 \times \frac{2}{2 + 4} = 4 \text{ A}$

Check: $5.33 + 2.67 + 4 = 12 \text{ A}$ (KCL). Faster in one line: the current in each branch is proportional to its **conductance** $1/R$, i.e. to $1/3 : 1/6 : 1/4 = 4 : 2 : 3$, so 12 A splits as $12 \times \frac{4}{9}$, $12 \times \frac{2}{9}$, $12 \times \frac{3}{9}$.

LABEL IN THE CLASS NOTES

The side note " $I_2 + I_3 = 6 \times 4 / 10 = 2.4$ " is the **resistance** $6 \parallel 4 = 2.4 \text{ k}\Omega$, not a current.

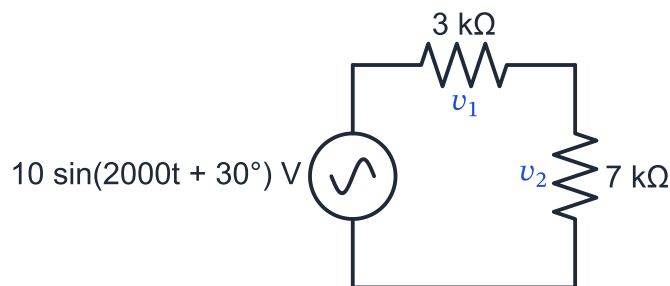
CHECK 2

(a) 10 mA into $1 \text{ k}\Omega \parallel 4 \text{ k}\Omega$: current in the $1 \text{ k}\Omega$? (b) Why does the *smaller* resistor take the *larger* current?

3 • Division works for AC too

CLASS P.5

$v = 10 \sin(2000t + 30^\circ) \text{ V}$ is applied across $3 \text{ k}\Omega$ and $7 \text{ k}\Omega$ in series. Find v_1 and v_2 .

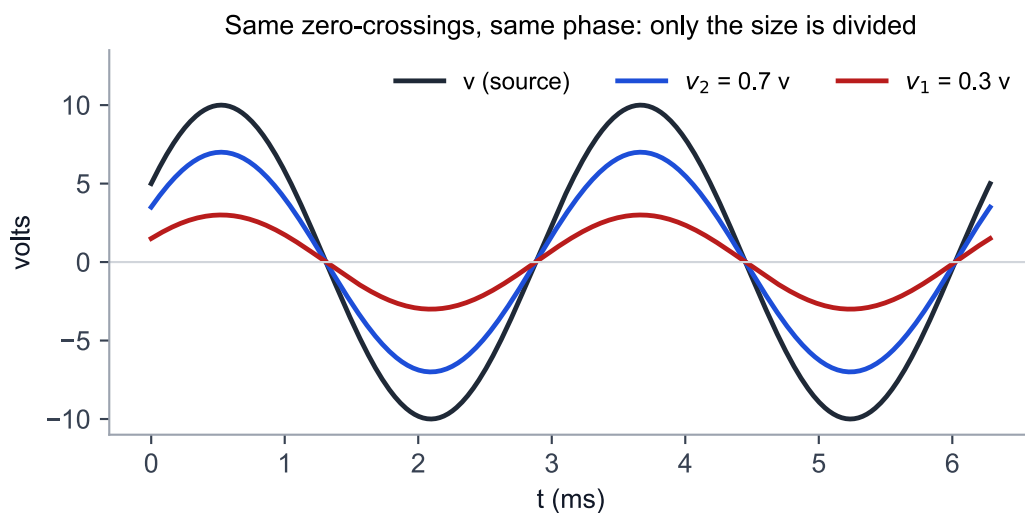


GUESS FIRST

Does v_1 have the same phase angle as the source, or a different one?

A resistor's $v = iR$ holds **at every instant**, so the divider ratio applies to the whole waveform. Resistors do not shift phase:

$$v_1 = 0.3 \times 10 \sin(2000t + 30^\circ) = 3 \sin(2000t + 30^\circ) \text{ V} \quad v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$$



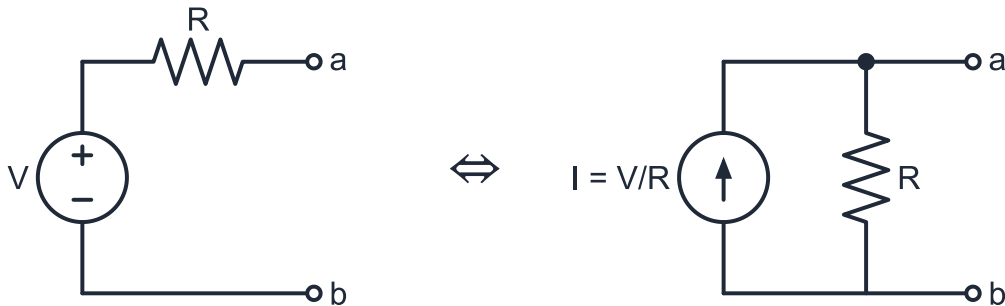
All three waves cross zero together: same frequency, same phase. Only the amplitude is divided.

This holds only because both elements are **resistors**. With an L or C in the divider, the ratio becomes complex and shifts the phase. That case uses impedances (05, conjugate matching).

4 · Source transformation

PREDICT

A 10 V source with 2 k Ω in series. Short its terminals: what current flows? Leave them open: what voltage appears? Could a current source with a resistor in parallel give the same two answers?



Both give the same open-circuit voltage (V) and the same short-circuit current (V/R), so no outside circuit can tell them apart.

A **voltage source V in series with R** is equivalent to a **current source $I = V/R$ in parallel with the same R** (class p.4), as seen from the terminals a–b.

For the Predict: short-circuit current = $10/2\text{k} = \mathbf{5\text{ mA}}$; open-circuit voltage = $\mathbf{10\text{ V}}$. So it is equivalent to 5 mA $\parallel$ 2 k Ω , and $5\text{ mA} \times 2\text{ k}\Omega = 10\text{ V}$, the same terminal behaviour.

It is the bridge between Thévenin and Norton in 04: $I_N = V_{Th} / R_{Th}$.

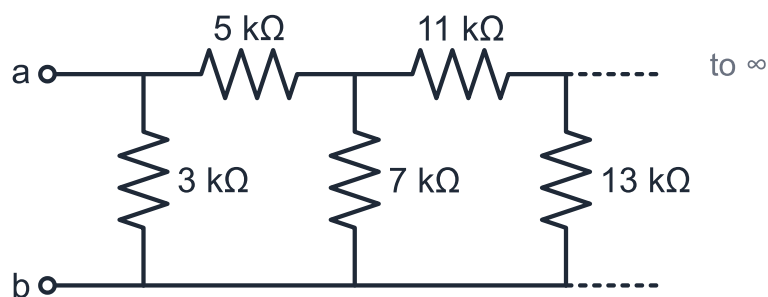
CHECK 3

Convert 3 mA in parallel with 4 k Ω into a voltage-source form.

5 · Series–parallel reduction: the infinite ladder

CLASS P.1 · THE FIRST QUESTION OF THE COURSE

Find R_{eq} of the infinite ladder: 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ... ∞ .



Read from the far end: each shunt sits in parallel with everything to its right; each series resistor adds.

GUESS FIRST

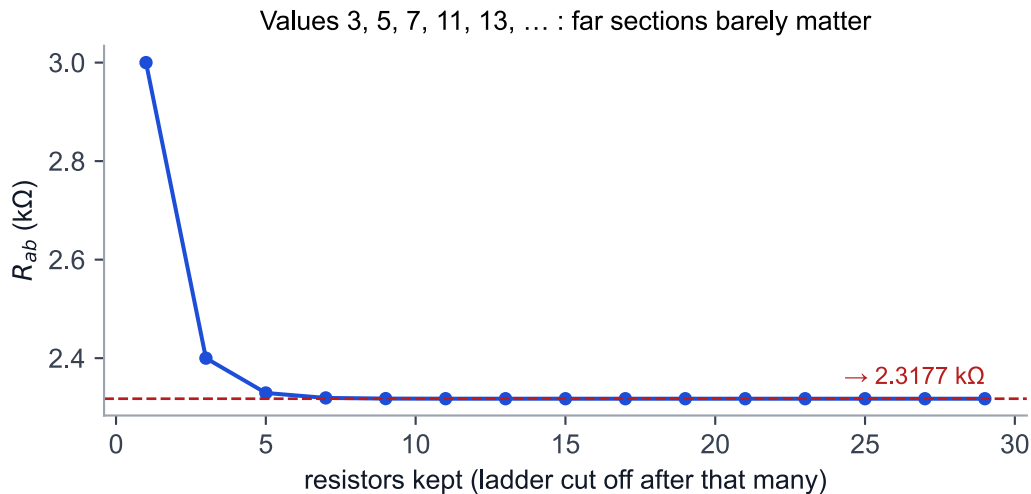
The first resistor is 3 k Ω , directly across a–b. Can R_{eq} be more than 3 k Ω ?

Rule: reduce from the far end towards the terminals. Written out, the ladder is:

$$R_{eq} = 3 \parallel (5 + 7 \parallel (11 + 13 \parallel (...))) \quad k\Omega$$

The Guess answer: the 3 kΩ is in parallel with everything else, so $R_{eq} < 3 \text{ k}\Omega$, always.

The values 3, 5, 7, 11, 13 are the **prime numbers**. Assuming that pattern continues (the notes stop at 13), cut the ladder off after more and more resistors:



3 resistors give 2.400 kΩ, 5 give 2.329, 9 give 2.318. The far sections are huge resistors hidden behind parallel paths, so they barely change the answer.

$R_{eq} \approx 2.318 \text{ k}\Omega$ (converged; confirmed by computer). The professor may intend a different continuation, so ask.

THE STANDARD TRICK, AND WHY IT FAILS HERE

For a ladder whose sections are **all identical** (series R, shunt R, repeated forever), cutting off the first section leaves the **same** infinite ladder. So $R_{eq} = R + (R \parallel R_{eq})$. Solving the quadratic gives $R_{eq} = R(1 + \sqrt{5})/2 \approx 1.618 R$.

Here every section is different, so the rest of the ladder never looks like the whole, and you must truncate.

Exam form

- **VDR:** $V_k = V \cdot \frac{R_k}{R_{total}}$. **CDR (two branches):** $I_1 = I \cdot \frac{R_2}{R_1 + R_2}$.
- Division holds for any waveform when every element is a resistor. Phase is unchanged.
- **Source transformation:** V in series with R $\Leftrightarrow$ V/R in parallel with R.
- **Ladder:** reduce from the far end. Identical infinite ladder: $R_{eq} = R + R \parallel R_{eq}$.

Traps

- CDR with the **same** resistor on top ($I \cdot R_1 / (R_1 + R_2)$ for I_1) is the most common slip: the answer comes out swapped.
- Source transformation keeps R **the same size**. It changes from series to parallel, not in value.
- Forgetting that R_{eq} of a network with a shunt resistor across the terminals must be smaller than that resistor.

Self-test

1. 20 V across 2 k Ω and 8 k Ω in series. Both voltages?
2. 9 mA into 3 k Ω || 6 k Ω . Both currents?
3. $v = 6 \sin(100t)$ V across 1 k Ω and 2 k Ω in series. v across the 2 k Ω ?
4. Transform 12 V in series with 3 k Ω .
5. An infinite ladder of identical 1 k Ω sections (series then shunt). R_{eq} ?

Answers

Check 1. $12 \times 2/6 = 4 \text{ V}$.

Check 2. (a) $10 \times 4/5 = 8 \text{ mA}$. (b) Same voltage across both, and $I = V/R$, so the smaller R takes more current.

Check 3. $3 \text{ mA} \times 4 \text{ k}\Omega = 12 \text{ V}$ in series with $4 \text{ k}\Omega$.

Self-test.

1. **4 V** and **16 V**.

2. $3 \text{ k}\Omega$: $9 \times 6/9 = 6 \text{ mA}$; $6 \text{ k}\Omega$: **3 mA**.

3. $\frac{2}{3} \times 6 \sin(100t) = 4 \sin(100t) \text{ V}$.

4. **4 mA in parallel with $3 \text{ k}\Omega$** .

5. $R(1 + \sqrt{5})/2 = 1.618 \text{ k}\Omega$.